MILITARY SPECIFICATION

ALBUMIN, NORMAL HUMAN SERUM, WITH INJECTION APPARATUS

This specification is mandatory for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 This specification covers normal human serum albumin and the injection apparatus necessary for use in administration.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

GG-I-526	Instruments, Dental and Surgical; General Specifications.
GG-N-196	Needles, Hypodermic, For Luer Syringes.
ZZ-T-831	Tubing; Rubber and Plastic.
MMM-A-250	Adhesive, Water-Resistant (For Sealing Fiberboard Boxes).
PPP-B-636	Box, Fiberboard.
PPP-C-96	Can, Metal, 28 Gage and Lighter.
PPP-C-186	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing).

STANDARDS

FEDERAL

Fed. Std. No. 1l2a	Parenteral Preparations.
Fed. Std. No. 595	Colors.

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN IRON AND STEEL INSTITUTE

Steel Products Manual, Stainless and Heat-Resisting Steels.

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East 42nd Street, New York, New York 10017.)

AMERICAN STANDARDS ASSOCIATION, INC.

ASA Z70.1-1955 (UDC615.47) American Standard Dimensions of Glass and Metal Luer Tapers for Medical Applications.

(Application for copies should be addressed to the American Standards Association, Inc., 10 East 40th Street, New York, New York 10016.)

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, U. S. Department of Health, Education, and Welfare, Washington, D. C. 20204.)

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE
NATIONAL INSTITUTES OF HEALTH

Public Health Service Regulations
Minimum Requirements (as applicable).

(Application for copies should be addressed to the National Institutes of Health, Bethesda, Maryland 20014.)

U. S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States.

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042.)

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3. REQUIREMENTS

3.1 Material. Shall be normal human serum albumin, suitable for use in the treatment of shock under field conditions, and as a diuretic, and the injection apparatus necessary for use in administration.

3.1.1 Glass. Bottle shall be made of clear glass conforming to type I, as described in the Pharmacopeia of the United States (U.S.P.) for injection containers.

3.1.2 Rubber.

3.1.2.1 Closure. Closure (stopper) for the glass bottle shall be made of fresh stock, high-grade, nonoxidizing rubber, suitable in all respects for stoppering commercial biological products having a high protein content. The rubber stopper shall be of such quality that it will not cause a precipitation or a degradation in the albumin solution. The stopper shall withstand autoclaving for 1 hour at 20 pounds pressure, and sterilization in 5 percent phenol for 20 minutes, without becoming tacky or showing any indication of color bleeding. All stoppers shall be properly treated prior to use to remove all bloom and other foreign material. Free sulfur content shall be not more than 0.3 percent.

3.1.2.2 Rubber tubing. Rubber tubing shall conform to ZZ-T-831c for type V, class 1, light wall rubber tubing, transluscent color only, 1/8 inch in inside diameter.

3.1.3 Needles.

3.1.3.1 Hose-hub needle. The cannula of the hose-hub needle shall be fabricated of type 304 AISI stainless steel and shall conform to QQ-N-196 for cannula style a or b. The cannula shall be seamless drawn, or welded and drawn tubing. The hub shall be of the hose-hub type. The hub shall by type 302, 303, or 304, of QQ-I-526, or nickel-plated brass suitable for the intended use. The needle shall be so designed as to pierce stopper cleanly without formation of particles or cores.

3.1.3.2 Recipient (intravenous) needle. The cannula of the recipient (intravenous) needle shall be fabricated of type 304 AISI stainless steel. The cannula shall be seamless drawn, or welded and drawn tubing. The hub shall be type 302, 303, or 304, of QQ-I-526, or nickel-plated brass suitable for the intended use.

3.1.3.3 Airway cannula. Airway cannula shall be fabricated of type 304 AISI stainless steel. The cannula shall be seamless drawn, or welded and drawn tubing.

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3.1.3.4 Additional needle requirements. In addition, the hose-hub needles, recipient (intravenous) needles, and airway cannulas shall comply with C-2, D-1, D-2, D-3, E-1, and E-2f, as specified in GG-N-196. The tests for "Stiffness" and "Elasticity" shall not be applicable to the hose-hub needles. Test E-2b shall only be applicable to the recipient (intravenous) needles.

3.1.4 Observation tube (needle adapter). Observation tube (needle adapter) shall be made of glass, or nylon or other suitable sterilizable plastic.

3.1.5 Suspension tape (sling). Suspension sling shall be constructed of cotton tape, or other suitable material, and shall be approximately 3/8 inch wide. The suspension sling shall have a dull, nonreflecting color, such as, dark gray, olive drab, or khaki.

3.1.6 Cotton. Cotton shall be clean, high grade, nonabsorbent cotton.

3.1.7 Filter for intravenous injection set. Filter for intravenous injection set shall be of stainless steel or nickel-copper alloy, and of suitable design for the purpose intended.

3.2 Components. The components (except packaging) for the unit of normal human serum albumin shall be as follows:

(1) One rubber-stoppered glass bottle containing Normal Human Serum Albumin.

(2) One suspension sling.

(3) One recipient (intravenous) needle.

(4) One airway cannula.

(5) One intravenous injection set (shall include one of each of the following: Hose-hub needle, filter, rubber tubing, observation tube (needle adapter), and end protectors).

(6) One instruction sheet.

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3.2.1 Normal human serum albumin. Shall be normal human serum albumin, U.S.P., and shall be in accordance with all applicable requirements of Fed. Std. No. 142a, together with the options and additions stated herein:

S2. Classification. Shall be type I, class 1.

Shall be suitable for intravenous use.

S5.2.1 The following additional requirements and tests are added to this paragraph:

Shall be a 25.0 percent w/v solution of Normal Serum Albumin (Human), within the applicable assay limits of the N.I.H.

Shall contain no preservative, but may contain stabilizing agents.

Shall be moderately viscous, clear, yellowish or brownish fluid; substantially odorless.

S6.3 Volume in final container. The final container (bottle) shall contain 100.0 cc, plus the required excess.

S6.8.1 In table I, Add to column No. 1, "120" and to column No. 2, "114."

3.2.2 Bottle. Bottle shall have an overflow capacity of 125.0 cc + 5 cc, and shall measure 1-3/4 inches to 2 inches in diameter and 4-1/4 inches ± 1/8 inch in height. The bottle finish shall be GCMI No. 20-2710 or No. 20A-2710. Graduation markings at 25-cc intervals shall be molded, etched, or baked on the bottle, or shall be printed on the edge of the label (see 5.2.1) in such a manner so that when the bottle is inverted, the graduations shall, at a glance, reveal the volume of liquid in the bottle during administration of the contents.

3.2.2.1 Label on the bottle shall bear the information specified in 5.2.1.

3.2.2.1.1 Clarity and permanence of markings. All required markings applied to the bottle shall be clear and legible, and the label shall remain entirely affixed to the bottle when tested as described in 4.4.4. All markings on all containers shall be durable and shall not smear or be removed during normal handling of the containers.

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3.2.2.2 Stopper. Stoppers shall be of material specified in 3.1.2.1, and shall be standard plug-type, rubber diaphragm stopper.

3.2.2.3 Seal. Stopper shall be secured to the bottle with corrosion-resistant aluminum, roll-on, tamperproof seal. The seal shall be placed in such a manner as to cover the stopper to prevent tampering with the contents.

3.2.3 Suspension tape (sling). Suspension tape shall be of sufficient length and so constructed as to serve as a means of suspending the bottle in an inverted position. The tape sling shall fit around the neck and body of the bottle in such a manner as to permit ready administration of the contents.

3.2.4 Recipient (intravenous) needle. Recipient (intravenous) needle shall be a 17-gage, 1-1/2 inch cannula hypodermic needle in accordance with the requirements of GG-N-196 for type I, point style b, hub style 1, or hub style 2, needle. The cannula and hub shall be fabricated of the material as specified in 3.1.3.2. The recipient (intravenous) needle shall fit the observation tube (needle adapter) specified in 3.2.6, paragraph (d), in such a manner as to make a tight leakproof connection therewith. Style 1 hubs may be fabricated from round stock, but shall be provided with a satisfactory finger grip (such as a knurled area) for attaching the needle to the observation tube (needle adapter). The hub cavities of both style 1 and style 2 hubs shall be constructed in accordance with figure 3 of ASA Standard Z70.1-1955. The hub cavities shall be tested for diameter and depth and fluidtightness as specified in 4.4.2 and 4.4.3. Hubs shall fit the plug gage within the "GO"—"NO GO" limits of the feeler gage as shown in figures 5 and 6 of ASA Standard Z70.1-1955. Hubs which do not fit the plug gage within these limits shall be considered not acceptable. In addition, when the hub is fitted onto the plug gage as specified in 4.4.2, there shall be no movement of the hub on the gage when the cannula is pressed lightly from side to side in all directions. When tested for fluidtightness, as specified in 4.4.3, the swaged connection of the cannula with the hub, and the connection of the hub, with the test gage, shall not show leakage exceeding the formation of a falling drop.

3.2.4.1 The entire cannula of the needle shall be enclosed in a plastic tubing sealed at one end. The needle and plastic tubing shall be further enclosed in an envelope which shall insure maintenance of sterility. If the recipient (intravenous) needle is packaged in a container which is not transparent, the container shall be marked to permit ready identification of the contents (i.e., 1-recipient (intravenous) needle)).

3.2.4.2 Sterility and freedom from pyrogens tests. The recipient (intravenous) needle shall be sterile and free from pyrogens when tested in accordance with 4.4.1.

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3.2.5 Airway cannula. The airway cannula shall be a properly pointed 16 gage cannula with a swaged-on bead or finger grip, approximately 1/4 inch in diameter, on the end opposite the bevel. The finger grip or bead shall be approximately 5/64 inch in diameter and approximately 1/4 inch long. The cannula shall be approximately 4 inches long and shall extend nearly to the bottom of the bottle when inserted through the stopper. The airway cannula shall extend above the level of the liquid when the bottle is inverted for use. The airway cannula shall be contained in a sleeve of clear plastic tubing closed at one end by heat-sealing, and at the other end with a cotton plug or glass bead. The cotton or bead shall be held securely in place, but shall be capable of being easily removed with the fingers. The sleeve shall be such that after sterilization, the sterility of the airway cannula shall be maintained.

3.2.5.1 Sterility and freedom from pyrogens tests. The airway cannula shall be sterile and free from pyrogens when tested in accordance with 4.4.1.

3.2.6 Intravenous injection set. The intravenous injection set shall consist of one (1) each of the following:

(a) Hose-hub needle, 16 gage, 3/4 inch, with cannula of the needle enclosed in plastic tubing (end protector), the free end of which shall be sealed. The plastic tubing shall be such that after sterilization, the sterility of the cannula shall be maintained. The hub of the needle shall be securely inserted into one end of the rubber tubing.

(b) Filter shall be approximately 100 mesh, and shall be located in the intravenous injection set.

(c) Rubber tubing shall be one (1) piece; or two lengths of tubing may be used if the filter which is employed necessitates the use of two lengths of tubing.

(d) Observation tube (needle adapter) shall be provided with a tapered tip, as specified below, and the other end shall have an outside diameter of 4 mm to 5 mm, and the overall length shall be 25 mm to 30 mm. The inside diameter shall be 1.5 mm to 3 mm. Shall be provided with a tapered tip to fit the recipient needle (intravenous needle) having a standard Luer hub in accordance with ASA Standard Z70.1-1955, in a tight, leakproof manner (see 3.2.4). The exposed portion of the needle adapter shall be covered with a 2-inch + 1/8 inch length of rubber tubing (end protector), "dog-eared" for ease in removal, open end of which shall have loosely packed nonabsorbent cotton to maintain the sterility of the tube and act as an air filter. The other end of the observation tube (needle adapter) shall be securely attached to the rubber tubing which shall cover 8 mm to 11 mm of the observation tube (needle adapter).

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3.2.6.1 Length of the intravenous injection set. The length of the unstretched injection set, measured from the bevel of the hose-hub needle to the Luer-type tip of the observation tube (needle adapter), shall be 43 inches ± 1 inch.

3.2.6.2 Sterility and freedom from pyrogens tests. The fluid path of the intravenous injection set, the cannula of the hose-hub needle, and the tip of the observation tube (needle adapter) shall be sterile and free from pyrogens when tested in accordance with 4.4.1. In addition, the fluid path shall be clean and free from foreign particles.

3.2.6.3 Safety test. The intravenous injection set shall meet the "Safety" test specified in the U.S.P. for "Transfusion and Infusion Assemblies."

3.2.7 Storage requirements. All glass bottles, after being filled with the normal human serum albumin and sealed, as specified, shall be stored continuously under constant refrigeration (between 2° - 8° C.), preferably at the lower limit. The filled bottles may be removed from the refrigerator(s) to complete the required packaging and packing which shall be accomplished within a minimum period of time.

3.3 Certificates. In addition to the certificates and the N.I.H. release required by Fed. Std. No. 142a, the contractor shall furnish the following with each delivery:

(a) A certificate of sterility for each component required to be sterile, listing all lot or control number, which shall state that samples from each sterility lot have been tested in accordance with N.I.H. method and found to be sterile.

(b) A certificate of freedom from pyrogens for each component required to be free from pyrogens, listing all lot or control numbers, which shall state that samples from each pharmaceutical lot have been tested in accordance with N.I.H. method and found to be free from pyrogens.

(c) A certificate of "Safety" for the intravenous injection sets, listing all lot or control numbers which shall state that samples from each lot have been tested in accordance with the U.S.P. method and found to comply.

Certificates and release shall be submitted for source and destination inspection as indicated in paragraph S9.1.1 of Fed. Std. No. 142a.

3.4 Workmanship. The material and its containers shall be free from any defects which detract from their appearance or may impair their serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibilities for inspection. Such examinations and tests as are set forth in this specification, or as shall otherwise be appropriate or necessary to insure that supplies conform to specification requirements, shall be performed by and at the expense of the supplier. Suppliers who do not have facilities adequate for such tests shall arrange for the use of test facilities acceptable to the Government. Records of examinations and tests performed by the supplier shall be maintained by the supplier and made available to the Government, upon the Government's request, at any time, or from time to time, for a period of 3 years after delivery of the supplies to which such records relate.

4.2 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

For visual examination	Inspection level	AQL (percent defective)	Unit of product
Major A	S-3	1.0	Can (with components)
Major B	S-3	1.0	Filled canister
Major C	S-3	1.0	Filled bottle
Major D	S-3	1.0	Needle (each type)
Major E	II	1.0	Intermediate package
Major F	II	1.0	Exterior package
Minor A	S-3	2.5	Can (with components)
Minor B	S-3	2.5	Needle (each type)

For end item testing:

Unit of product	Characteristic	Requirement paragraph	Sampling level
Filled bottle:			
	Label adhesion	3.2.2.1.1	S-2 (minimum of 10)
Normal Human Serum Albumin	U.S.P. Monograph Tests (N.I.H. Release Tests)	3.2.1	U.S.P. procedures U.S.P.H.S. Regulations
	Volume	3.2.1	U.S.P. procedures
(Glass)	Chemical resistance	3.1.1	Sufficient to yield 100 Gm.
	Longitudinal strain		S-2; AQL 4.0

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Unit of product	Characteristic	Requirement paragraph	Sampling level
(Rubber stopper)	Resistance to autoclaving	3.1.2.1	S-2; AQL 2.5
	Phenol sterilization		
	Free sulfur		
Filter	Composition	3.1.7	Sufficient to yield 3 ounces
Needle cannulas	Corrosion resistance	3.1.3.4	S-3; AQL 4.0
	Composition	3.1.3.1, 3.1.3.2 and 3.1.3.3	Sufficient to yield 3 ounces
	Security of swaging	3.1.3.4	S-1; AQL 2.5
	Stiffness	3.1.3.4	S-1; AQL 4.0
	Elasticity	3.1.3.4	S-1; AQL 4.0
	Resistance to breakage	3.1.3.4	S-1; AQL 2.5
Needle hub	Corrosion resistance	3.1.3.4	S-3; AQL 4.0
	Foreign matter	3.1.3.4	S-2; AQL 1.5
Needles (Airway cannula and recipient (intravenous)	Sterility	3.2.4.2	U.S.P. Procedure
	Freedom from pyrogens	3.2.5.2	U.S.P. Procedure
Injection assembly Hose hub needle, Cannula, and Observation tube (needle adapter) fluid path	Sterility and freedom from pyrogens	3.2.6.2	U.S.P. Procedure
	Safety	3.2.6.3	U.S.P. procedure
Can	Material and construction	5.3.3.1.4	
	Material and coating	5.3.3.1.4.3	PPP-C-96
	Protective coating	5 3.3.1.4	
Rubber tubing (1 yard length)	Specific gravity	3.1.2.2	
	Tensile stress	Fed. Spec. ZZ-T-831b	Fed. Test Method Std. No. 601
	Tensile strength	ditto	
	Ultimate elongation	ditto	
	Accelerated aging	ditto	
	Heavy metals		

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Unit of product	Characteristic	Requirement paragraph	Sampling level
Cotton	General	3.1.6	----
Suspension tape	General	3.1.5	----
Observation tube (needle adapter)	Composition	3.1.4	----
Tubing (plastic) for protecting needles	Composition	---	----

4.2.1 Army Medical Research samples. The quality assurance representative, when inspection is at source, or the contractor, when inspection is at destination, shall send the following by fastest route directly to:

Blood Transfusion Division
Army Medical Research Laboratory
Fort Knox, Kentucky 40121

(1) One (1) unit (including circular) from each lot or batch, accompanied by appropriate document bearing the following information:

(a) Federal stock number
(b) Item identification
(c) Contract number
(d) Contractor's name and address
(e) Lot or control number
(f) Quantity of material represented by the sample.

(2) One (1) copy of the protocol which the manufacturer has submitted to N.I.H., stating the essential manufacturing processes and results of test for each lot or batch.

(3) One (1) copy of N.I.H. release for each lot or batch.

Shipment of material shall not be withheld for a report from Army Medical Research Laboratory.

4.3 Examination. The normal human serum albumin and the components shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.2.

4.3.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

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Categories and defects

Major A

101. Finger grip not affixed in proper location.
102. Labeling not correct, not complete, not legible.
103. Any required component not within can (and protective bags).
104. Interior of can not cushioned correctly.
105. Closure of protective bag not as described.
106. Excess air not expelled from bag.
107. Top of can not suitably scored to provide complete removal.

Major B

121. Injection set coiling, when uncoiled, not free of kinks.
122. Clean cotton not inserted at open end of rubber tube covering observation tube (needle adapter).
123. Hub of injection assembly (hose hub) needle and observation tube (needle adapter) not securely inserted into opposite ends of rubber tubing.
124. Rubber tubing and plastic covers not free of pinholes.
125. Filter not in injection assembly.

Major C

131. Solution not clear yellowish or brownish, not free of turbidity.
132. Solution not free of particulate matter.
133. Graduation markings not present.
134. Label not complete, not legible.
135. Stopper not standard plug type rubber diaphragm.
136. Metal seal on bottle not roll-on and tamperproof.
137. Stopper not completely covered by metal seal.
138. Suspension sling not sufficient length, not in proper position.
139. Recipient (intravenous) needle not secured to bottle.

Major D

141. Hub and cannula outer surface not smoothly finished and free of pits and tool marks.
142. Cannula inner passage not clear and free of scale, deep pits, deep furrows or die marks.
143. Point not keen and properly formed or free of burs and slivers.
144. Recipient, airway cannula, and injection assembly hub type not correct gage.
145. Recipient (intravenous) needle not encased at cannula end with plastic tubing.
146. Plastic tubing for recipient (intravenous) needle not sealed at one end.
147. Covered recipient (intravenous) needle not enclosed in envelope.
148. Recipient (intravenous) needle envelope not transparent or identified.
149. Airway cannula not enclosed in a clear plastic tubing.
150. Airway cannula plastic cover not heat sealed at one end.
151. Injection assembly hub needle cover not heat sealed.

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Major E

181. Packaging quantities not correct.
182. Full height partitions not present.
183. Container not properly taped.
184. Container not W5c board.
185. Marking not as specified.

Major F

191. Packaging quantities not correct.
192. Container not V3c board.
193. Container not properly taped.
194. Marking not as specified.

Minor A

201. Complete injection assembly not 43 inches ± 1 inch.
202. Inside diameter of rubber tubing not as specified.
203. Suspension tape not dull, nonreflecting color.
204. Suspension tape not approximately 3/8 inch wide.
205. Observation tube (needle adapter) not dimensions specified.
206. Observation tube (needle adapter) tip not standard Luer-type.
207. Rubber tubing covering observation tube (needle adapter) tip not 2 inches "dog-eared".
208. Bottle overflow capacity not 125 cc ± 5 cc.
209. Bottle dimensions not as specified.
210. Recipient (intravenous) needle securing tape not free of label.
211. Bottle not in closure end-up position when in can.
212. Canister not dimensions specified.
213. Outside dimensions of can not as specified.
214. Graduation markings not as specified.
215. Graduation markings not completely visible.

Minor B

251. Airway cannula not proper length (approximately 4 inches).
252. Injection assembly hub type needle not 3/4 inch in length.
253. Recipient (intravenous) needle not 1-1/2 inches in length.

4.4 Tests shall be conducted in accordance with the applicable specifications, as necessary, to determine compliance with this specification.

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4.4.1 Needles, airway cannula, and intravenous injection set. The needles, airway cannula, and intravenous injection set shall be tested for sterility and freedom from pyrogens in accordance with the requirements of the N.I.H. In addition, the intravenous injection set shall meet the "Safety" test required herein.

4.4.2 Diameter and depth of hub cavity. The needle hub cavity shall be tested on the male Luer taper gage in accordance with plug gage of figures 5 and 6 of ASA Standard Z70.1-1955. (This gage shall be as shown in ASA Standard Z70.1-1955, except that it shall be constructed with a suitable fluid path to permit performance of the fluidtightness test specified in 4.4.3). The hub shall be slipped onto this gage with very light pressure. The fit of the hub on this gage shall be tested by means of the feeler gage in accordance with figures 5 and 6 of ASA Standard Z70.1-1955.

4.4.3 Fluidtightness. With the needle in a horizontal position, the hub shall be attached by means of a plug gage, as shown on figure 5 of ASA Standard Z70.1-1955 (except that gage shall have a fluid path), to a water pressure source having a pressure gage. The water shall be run through the needle so as to fill any hub cavity space and the lumen of the cannula; the lumen then sealed by a piece of solid rubber slipped on to the point and a pressure of 45 pounds per square inch applied for 10 seconds.

4.4.4 Adhesion of labels and durability of markings on the labels. Five (5) to ten (10) labeled immediate containers of Albumin, Normal Human Serum, U.S.P., 100 cc, shall be selected at random from each lot. The containers shall be placed in a bath of water maintained at a temperature of 75° F. + 5° F. for a period of 4 hours, during which time this label shall be completely covered by the water. When the containers are removed from the water, the labels shall be entirely affixed to the containers, and all the markings on the labels shall be clear and legible. After drying, the labels shall remain entirely affixed to the containers and the markings on the labels shall be clear and legible.

4.4.5 Leak test. In the applicable FSC 6505 and 6810 Amendment No. 10, dated 12 January 1967, in paragraph 4.3.1.1, on page 2, add the following: "As an alternate method to the leak test specified, the manufacturer shall test all filled Normal Human Serum Albumin units for leakers as follows:

> Submerge the bottles in a water bath for ten (10) hours at 60° C. Remove and dry the bottles. Incubate the bottles at 25° C. for two (2) weeks. At the end of the incubation period, the bottles shall be 100 percent inspected for detectable leakers.

For inspection purposes, the Government may, at its discretion, test five (5) filled Normal Human Serum Albumin units in accordance with the leakage test specified in paragraph 4.3.1.1. None of the five (5) units tested shall leak."

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4.4.6 Clear passage test. Samples, after sterilization, shall be subjected to an internal pressure of 3 pounds p.s.i. and, when subjected to this pressure, shall allow free passage of the testing fluid.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with the applicable paragraphs of PPP-C-186, and as specified herein.

5.2 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug, and Cosmetic Act and the National Institutes of Health, and shall include the following information:

5.2.1 Immediate container (bottle). Each immediate container (bottle) specified in 3.2.2, shall bear the following information:

(a) the expiration date

(b) the lot or control number

(c) the following statements:

1. Contains no preservative.

2. Salt Poor.

3. Store between 2° - 8° C. (35° - 46° F.).

4. Caution: Do not use if particulate matter, discoloration, sediment, or defective closure is present.

5. Caution: Federal law prohibits dispensing without prescription.

(d) the name and address of the manufacturer. When the manufacturer is not the contractor, the name and address of the contractor shall also appear. When both names are placed on the label, the following designations shall precede the names:

"MFR" for the manufacturer and
"CONTR" for the contractor

(e) graduation marks at 25-cc intervals at the label edge, if the bottle does not bear 25-cc graduations molded, etched, or baked thereon (see 3.2.2).

(f) the U.S. Government license number.

In addition to the above, each immediate container shall bear a label containing the following information and instructions for use:

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"ALBUMIN, NORMAL HUMAN SERUM

25 Gm in 100 cc of buffered diluent

CAUTION: DO NOT USE IF TURBID

INSTRUCTIONS FOR USE

1. Apply alcohol or iodine to rubber stopper.
2. Hold container in the upright position, insert airway cannula through rubber stopper so that pointed end of cannula reaches nearly to bottom of bottle.
3. Insert short needle of the intravenous set through the rubber stopper.
4. Attach recipient needle to adapter on the administration tubing.
5. Suspend container in inverted position approximately 3 feet above patient.
6. Allow tubing to fill with albumin solution.
7. Insert needle into vein.
8. In the presence of dehydration, albumin should be given with or followed by addition of fluids."

5.2.2 Unit packages (cans). Each unit package (can) shall be durably and legibly marked by lithographing with the following information. However, the information is not required to appear in the sequence indicated:

(a) the item identification designated as "ALBUMIN, NORMAL HUMAN SERUM, U.S.P."

(b) the quantity of contents per immediate container (bottle) designated as "100 cc"

(c) the stock number designated as "FSN 6505-299-8179" or "Stock No. 6505-299-8179"

(d) the lot or control number

(e) the date of manufacture

(f) the expiration date

(g) a list of the contents

(h) instructions for use

(The INSTRUCTIONS FOR USE shall be exactly as shown in 5.2.1 for the immediate containers and include the statements No. 1 through No. 8, inclusive).

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(i) the name and address of the manufacturer. When the manufacturer is not the contractor, the name and address of the contractor shall also appear.

When both names are placed on the label, the following designations shall precede the names:

"MFR" for the manufacturer and
"CONTR" for the contractor.

(j) the following legends:

"STORE BETWEEN 2° - 8° C. (35° - 46° F.)."

"DO NOT PERMIT TO FREEZE."

"DO NOT USE IF TURBID."

"CONTAINS NO PRESERVATIVE."

"Salt Poor."

"Caution: Federal law prohibits dispensing without prescription."

(k) the U. S. Government license number

(l) any other information required by the N.I.H and the U.S.P.

With the bottom of the can resting on a horizontal surface, the labeling information (marking) shall be in a horizontal position on the can.

5.2.3 Instruction sheet. Instruction sheet shall conform to the following, and shall be placed within each unit package (can).

INSTRUCTION SHEET

INSTRUCTIONS FOR THE CLINICAL USE OF NORMAL HUMAN SERUM ALBUMIN

CONTENTS

Each bottle contains 25 grams of normal human serum albumin in 100 cc of solution, prepared according to the Minimum Requirements of the National Institutes of Health, containing 0.04M Sodium Acetyltryptophanate or 0.02M Sodium Acetyltryptophanate with 0.02M Sodium Caprylate. This is the osmotic equivalent of approximately 500 cc of citrated plasma. The sodium acetyltryptophanate improves the stability of the solution and is a derivative of an amino acid in which human serum albumin is deficient. The solution contains no preservative. It has been sterilized by filtration.

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ACTION

When injected intravenously, 100 cc of 25 percent albumin draws approximately 350 cc of additional fluid into the circulation within 15 minutes, except in the presence of marked dehydration. This extra fluid reduces hemoconcentration and blood viscosity. If hemorrhage has occurred, an anemia may follow hemodilution. This should be treated with whole blood. When the circulating blood volume has been depleted, the hemodilution following albumin persists many hours. In individuals with a normal blood volume, it usually lasts only a few hours.

The low sodium content of Normal Human Serum Albumin is of value when large amounts are administered to hypoproteinemic patients with edema and poor kidney function, as the accumulation of sodium ion in the extracellular fluid is minimized. Normal Human Serum Albumin may be given in conjunction with or combined with other parenteral fluids -- whole blood, plasma; saline, glucose, or sodium lactate. The addition of two bottles of albumin to 1000 cc of normal saline or 5 percent glucose gives a solution approximately isotonic and isosmotic with citrated plasma.

INDICATIONS AND DOSAGE

Shock. Albumin is indicated in the emergency treatment of shock due to trauma, hemorrhage, operation, and infection, or in the treatment of injuries of such severity that shock, although not immediately present, is likely to ensue. An initial dose of 1 to 2 bottles can be followed in 15 to 30 minutes by a third bottle if the response is inadequate. Administration of larger quantities should be supplemented with or replaced by the use of whole blood to combat the relative anemia, which would follow the use of three bottles of albumin (equivalent to 1500 cc plasma). The quick response of blood pressure, which may follow the rapid administration of concentrated albumin, necessitates careful observation of the injured patient to detect bleeding points which failed to bleed at a lower blood pressure. Because the bottle of concentrated albumin can be administered in 3 to 10 minutes, it is preferable for the earlient treatment of shock. If dehydration is present, the administration of other fluids by any available route is desirable, either with albumin (two bottles of albumin per liter solution) or after albumin. Even in marked dehydration, however, concentrated albumin is not harmful. It is merely less effective.

Burns. An initial dose of 1 to 2 bottles should be followed by 5 percent solution of albumin (two bottles per liter) in normal saline or 5 percent glucose in such quantities as are needed to prevent marked hemoconcentration, and to maintain appropriate electrolyte balance.

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Hypoproteinemia with or without edema. Two to three bottles daily over a period of days or weeks may be required to reduce edema and to raise the serum protein values to normal. Since these patients have approximately normal blood volumes, doses of more than one bottle of albumin should not be given faster than 100 cc in 30 to 45 minutes to avoid circulatory embarrassment. If slower administration is desired, in cases with hypertension or cardiac failure, 200 cc of albumin solution may be mixed with 300 cc of 10 percent glucose solution in water and administered by continuous drip at a rate of 10 Gm of albumin (100 cc) per hour. Although diuresis may occur soon after albumin administration has been instituted, best results are obtained if albumin is continued until normal serum protein and serum albumin levels are regained.

IF THE SOLUTION IS TURBID, DO NOT USE.

5.3 Packaging.

5.3.1 Unit of issue. One (1) can, with components, as specified in 3.2, constitutes one unit of issue.

5.3.2 Packaging quantities. The number of units of issue indicated in table I shall be packaged in each unit, intermediate, and exterior container, as spplicable, for the required level of protection specified in the procurement document.

Table I. Packaging quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	48 units

5.3.2.1 Packing variation permitted. If the required number of uni in the entire shipment is less than the number of units specified to be over-packed in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.3.3 Level A.

5.3.3.1 Unit package.

5.3.3.1.1 Canister. The injection set shall be coiled, in a manner as to prevent kinking of the tubing, and placed in a canister fabricated of spirally-wound chipboard, with paper caps. The approximate size of the canister shall be 2 inches in diameter and 1-3/8 inches in height. The bottom cap shall have a die-cut hole of approximately oneinch, located in center of cap, and this cap shall be glued in position. The injection assembly shall be wound in a manner to form an open core which will permit placing the packaged canister around the neck of bottle.

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5.3.3.1.2 Protective bag. Protective bag shall be a bag fabricated from virgin polyethylene tubing of 0.004-inch average thickness, with no wall thickness at any point below 0.00325 inch. Excess air shall be expelled from bag and closure effected by heat sealing, or twisting open end of bag and securing with metal tie or cord. If closure is effected by heat sealing, a V-shaped notch shall be cut in bag above the closure to permit ready opening of bag. Required components, as specified in 3.2, shall be placed within protective bag.

5.3.3.1.3 Cushioning. Component parts shall be protected by cushioning within the can. A die-cut disc of double-faced corrugated fiberboard shall be placed in bottom of can. A sheet of single-faced, corrugated fiberboard or indented interior pulpboard shall be used to line the side walls of the can. The work shall be accomplished in such a manner as to prevent shifting of the contents, or damage to the component parts.

5.3.3.1.4 Metal can. The metal can used for packaging of component parts shall conform to the applicable requirements of PPP-C-96.

5.3.3.1.4.1 Type. Shall be type III in accordance with 3.3.3 of PPP-C-96, except that reclosure feature will not be required. Can shall be furnished with an aluminum pull type end, in lieu of a key opening band. The entire top of can shall be removable when tear tab is pulled.

5.3.3.1.4.2 Size. The outside dimensions of can shall not exceed 2-3/4 inches in diameter and 5-3/4 inches in height.

5.3.3.1.4.3 Metallic coating. Shall be 0.50 electrolytic tinplate of 1.25 hot dipped tinplate (see 3.1.1.1.1, 3.1.1.1.2, and 3.1.1.5 of PPP-C-96.)

5.3.3.1.4.4 Protective coatings.

5.3.3.1.4.4.1 Interior coating. Interior coating will not be required.

5.3.3.1.4.4.2 Exterior coating. The exterior surfaces of each can shall be coated in accordance with plan B, specified in 3.2.2.2 of PPP-C-96, except that the color shall conform to Color No. X34087 of Fed. Std. 595.

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5.3.3.1.5 Assembly. The component parts, specified in 3.2, shall be assembled and packaged as follows:

(a) Suspension sling shall be placed around bottle in a manner as to permit ready dispensing of the serum.

(b) Recipient (intravenous) needle, protected as specified in 3.2.4, and the airway cannula, protected as specified in 3.2.5, shall be secured to one side of the bottle with transparent pressure-sensitive tape. The work shall be accomplished in such a manner as to prevent obliteration of label markings when tape is removed.

(c) Injection set. The injection set, packaged as specified in 5.3.3.1.1, shall be positioned around neck of bottle, with the bottom of the canister resting on shoulder of bottle.

(d) Instruction sheet. Instruction sheet, in accordance with 5.2.3, shall be positioned around bottle.

(e) Protective bag. Component parts, positioned as specified above, shall be packaged in a polyethylene bag as specified in 5.3.3.1.2.

(f) Cushioning. Cushioning, as specified in 5.3.3.1.3, shall be positioned in can.

(g) Canning. The protected component parts shall be placed in can, properly cushioned, with the top of the bottle corresponding to the top (tear-tab end) of the can.

(h) Sealing of can. Air shall be replaced with dry nitrogen and sealed.

5.3.3.2 Intermediate package. Twelve cans (12 units) shall be packaged in a carton of appropriate size constructed in accordance with PPP-B-636, type CF, class weather-resistant. Box design shall include full-height partitions fabricated of the same material as the box. The partitions shall form an individual snug-fitting cell for each can. Closure shall be effected by taping the top and bottom flaps with 2 inch wide tape conforming to PPP-T-76. Closing of bottom flaps with adhesive conforming to MMM-A-250, or metal stitches is permitted.

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5.4 Packing.

5.4.1 Level A.

5.4.1.1 Exterior container. Exterior container shall be a box of appropriate size constructed in accordance with PPP-B-636, type CF, class weather resistant, grade V3c, style RSC.

5.4.1.2 Closure. Closure and strapping shall be as specified in the appendix of PPP-B-636.

5.5 Marking.

5.5.1 Unit package. Each can shall be marked in accordance with 5.2.2.

5.5.2 Intermediate package. Each intermediate package shall be labeled with the stock number, item identification, unit, quantity, lot or control number, contract number, date of manufacture (month, year), expiration date (month, year), name and address of contractor, and any such other data as may be required by the N.I.H. Regulations for Labeling of Biologicals. Marking shall include the legends:

"STORE BETWEEN 2° - 8° C. (35° - 46° F.)."
"DO NOT PERMIT TO FREEZE."

5.5.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Lot (control) numbers shall be shown. Date of manufacture and expiration date shall be shown in lieu of date packed. Precautionary markings as specified in MIL-STD-129 shall be used to indicate top of exterior container. In addition, each exterior container shall bear the legends:

"STORE BETWEEN 2° - 8° C. (35° - 46° F.)."
"DO NOT PERMIT TO FREEZE."

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5.6 Special labeling.

5.6.1 For refrigerated shipment. Each refrigerated exterior container shall be marked as indicated in paragraph 5.5.14 of MIL-STD-129.

5.6.2 For unrefrigerated shipment. Each unrefrigerated exterior container shall be marked in accordance with 5.5.15 of MIL-STD-129.

5.6.2.1 The unrefrigerated shipping time for this item shall be not more than 144 hours.

5.6.3 Shipping instructions. If unrefrigerated shipment is used, the item covered by this procurement document shall be forwarded by surface or air transportation which will insure receipt of the shipment by the consignee in less than six (6) days (144 hours), from the time the material was removed from refrigeration. When it is anticipated that the intransit period will exceed six (6) days, such shipments may be forwarded in refrigerated containers which will maintain the required temperature range of 35° to 46° F. while enroute to consignee. When refrigerated transportation is available from the consignor to the consignee, such transportation may be used in lieu of refrigerated containers.

5.6.4 Scheduling of shipments. Shipments should be scheduled to arrive at destination on a working day. Advance notice should be expeditiously forwarded to the consignee regarding the shipment and the estimated time of arrival.

6. NOTES:

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the type, etc., which is normally procured to meet Military requirements.

Custodians:
Army-MD
Navy-MS
Air Force-03

Review activities:
Army-MD
Navy-MS
Defense Supply Agency-DM

Preparing activity:
Defense Supply Agency-DM

Review information is current, as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements .

SPECIFICATION

MIL-A-36561

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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343-218 (A-218)

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